

## LAB # 12

### IMPLEMENTATION OF STRUCTURES IN C++

#### Lab Tasks

##### 1- Tasks

Create a structure to specify data of customers in a bank. The data to be stored is: Account number, Name, Balance in account. Assume maximum of 200 customers in the bank. Write a function to print the Account number and name of each customer with balance below Rs. 1000.

##### 2- Tasks

Write a menu driven program that depicts the working of a library. The menu options should be:

- a) Add book information
- b) Display book information
- c) List all books of given author
- d) List the title of specified book
- e) List the count of books in the library
- f) List the books in the order of accession number
- g) Exit

Create a structure called library to hold accession number, title of the book, author name, price of the book, and flag indicating whether book is issued or not.

##### 3- Tasks

- i- Declare a structure named employee that stores the employee id, name, salary and department.
- ii- Declare an array of 5 employees for the structure defined in part(i) . Also write statements to assign the following values to the employee [3].  
**Employee id = “Your\_roll\_no” salary = 30,000 and department = “IT dept”**
- iii- Write necessary statement to initialize all the elements of above array.
- iv- Write a function to take input in above array of struct employee.
- v- Write a function that prints the highest salaried person amongst the employees.
- vi- Write a function that search & display records of all those employees, whose salary is greater than 15000.

- vii- Write a function that search & display records of all those employees, who are working in Finance department.

## 1- Tasks

Create a structure to specify data of customers in a bank. The data to be stored is: Account number, Name, Balance in account. Assume maximum of 200 customers in the bank. Write a function to print the Account number and name of each customer with balance below Rs. 1000.

```
#include <iostream>
using namespace std;

struct customer {
    int Acc_num;
    string name;
    double balance;
}c[200];

void bank() {
    for (int i = 0; i < 3; i++) {
        cout << "Enter Account Number: ";
        cin >> c[i].Acc_num;
        cout << "Enter Name: ";
        cin >> c[i].name;
        cout << "Enter Balance: ";
        cin >> c[i].balance;
        cout << endl;
    }
    cout << "Customers with balance below 1000"<<endl;
    for (int j = 0; j <3 ; j++) {
        if (c[j].balance < 1000) {
            cout << "Name: " << c[j].name<<endl;
            cout << "balance: " << c[j].balance<<endl;
        }
    }
}

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023\n";
    bank();
}
```

```
Muhammad Ali
2025F-BIT-023
Enter Account Number: 1
Enter Name: ali
Enter Balance: 1500

Enter Account Number: 2
Enter Name: ahmed
Enter Balance: 500

Enter Account Number: 3
Enter Name: umar
Enter Balance: 2000

Customers with balance below 1000
Name: ahmed
balance: 500
```

## 2- Tasks

Write a menu driven program that depicts the working of a library. The menu options should be:

- a. Add book information
- b. Display book information
- c. List all books of given author
- d. List the title of specified book
- e. List the count of books in the library
- f. List the books in the order of accession number
- g. Exit

Create a structure called library to hold accession number, title of the book, author name, price of the book, and flag indicating whether book is issued or not.

```

    #include <iostream>
    #include <string>
    using namespace std;

    struct library {
        int accession_num;
        string title;
        string author;
        double price;
        string flag;
    };

    int main() {
        cout << "Muhammad Ali\n2025F-BIT-023\n\n";

        library books[100];
        int count = 0;
        int choice;

        do {
            cout << "LIBRARY MENU\n";
            cout << "1. Add book information\n";
            cout << "2. Display book information\n";
            cout << "3. List all books of given author\n";
            cout << "4. List the title of specified book\n";
            cout << "5. List the count of books in the library\n";
            cout << "6. List the books in order of accession number\n";
            cout << "7. Exit\n";
            cout << "Enter your choice: ";
            cin >> choice;

            switch (choice) {
                case 1:
                    cout << "Enter Accession Number: ";
                    cin >> books[count].accession_num;
                    cin.ignore();

                    cout << "Enter Book Title: ";
                    getline(cin, books[count].title);

                    cout << "Enter Author Name: ";
                    getline(cin, books[count].author);

```

```
        cout << "Enter Price: ";
        cin >> books[count].price;
        cin.ignore();

        cout << "Issued or Not Issued: ";
        getline(cin, books[count].flag);

        count++;
        cout << "Book information added\n";
        break;

    case 2:
        for (int i = 0; i < count; i++) {
            cout << "\nBook " << i+1 << endl;
            cout << "Accession No: " << books[i].accession_num << endl;
            cout << "Title: " << books[i].title << endl;
            cout << "Author: " << books[i].author << endl;
            cout << "Price: " << books[i].price << endl;
            cout << "Status: " << books[i].flag << endl;
        }
        break;

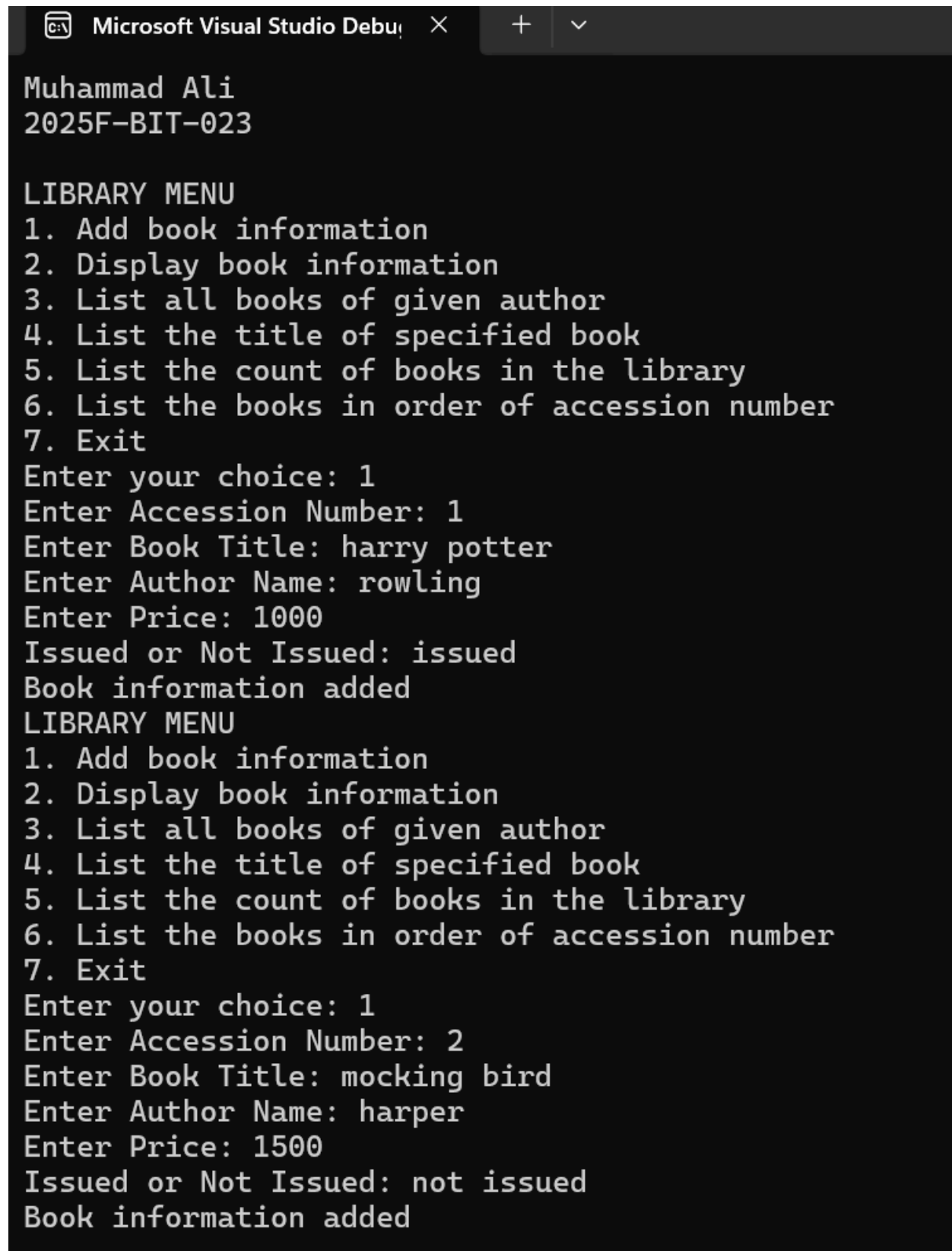
    case 3: {
        string author;
        cout << "Enter author name: ";
        cin.ignore();
        getline(cin, author);

        for (int i = 0; i < count; i++) {
            if (books[i].author == author) {
                cout << books[i].title << endl;
            }
        }
        break;
    }

    case 4: {
        int acc;
        cout << "Enter accession number: ";
        cin >> acc;

        for (int i = 0; i < count; i++) {
```

```
        for (int i = 0; i < count; i++) {  
            if (books[i].accession_num == acc) {  
                cout << "Title: " << books[i].title << endl;  
            }  
        }  
        break;  
    }  
  
    case 5:  
        cout << "Total books in library: " << count << endl;  
        break;  
  
    case 6:  
        for (int i = 0; i < count - 1; i++) {  
            for (int j = i + 1; j < count; j++) {  
                if (books[i].accession_num > books[j].accession_num) {  
                    library temp = books[i];  
                    books[i] = books[j];  
                    books[j] = temp;  
                }  
            }  
        }  
  
        cout << "\nBooks list:\n";  
        for (int i = 0; i < count; i++) {  
            cout << books[i].accession_num << " | " << books[i].title << endl;  
        }  
        break;  
  
    case 7:  
        cout << "Exit\n";  
        break;  
  
    default:  
        cout << "Invalid choice!\n";  
    }  
} while (choice != 7);  
  
return 0;  
}
```



```
Microsoft Visual Studio Debug Console
Muhammad Ali
2025F-BIT-023

LIBRARY MENU
1. Add book information
2. Display book information
3. List all books of given author
4. List the title of specified book
5. List the count of books in the library
6. List the books in order of accession number
7. Exit
Enter your choice: 1
Enter Accession Number: 1
Enter Book Title: harry potter
Enter Author Name: rowling
Enter Price: 1000
Issued or Not Issued: issued
Book information added
LIBRARY MENU
1. Add book information
2. Display book information
3. List all books of given author
4. List the title of specified book
5. List the count of books in the library
6. List the books in order of accession number
7. Exit
Enter your choice: 1
Enter Accession Number: 2
Enter Book Title: mocking bird
Enter Author Name: harper
Enter Price: 1500
Issued or Not Issued: not issued
Book information added
```



```
Microsoft Visual Studio Debu  X  +  v

LIBRARY MENU
1. Add book information
2. Display book information
3. List all books of given author
4. List the title of specified book
5. List the count of books in the library
6. List the books in order of accession number
7. Exit
Enter your choice: 2

Book 1
Accession No: 1
Title: harry potter
Author: rowling
Price: 1000
Status: issued

Book 2
Accession No: 2
Title: mocking bird
Author: harper
Price: 1500
Status: not issued
LIBRARY MENU
1. Add book information
2. Display book information
3. List all books of given author
4. List the title of specified book
5. List the count of books in the library
6. List the books in order of accession number
7. Exit
Enter your choice: 3
Enter author name: harper
```



Microsoft Visual Studio Debug Console



```
mocking bird
LIBRARY MENU
1. Add book information
2. Display book information
3. List all books of given author
4. List the title of specified book
5. List the count of books in the library
6. List the books in order of accession number
7. Exit
Enter your choice: 4
Enter accession number: 1
Title: harry potter
LIBRARY MENU
1. Add book information
2. Display book information
3. List all books of given author
4. List the title of specified book
5. List the count of books in the library
6. List the books in order of accession number
7. Exit
Enter your choice: 5
Total books in library: 2
LIBRARY MENU
1. Add book information
2. Display book information
3. List all books of given author
4. List the title of specified book
5. List the count of books in the library
6. List the books in order of accession number
7. Exit
Enter your choice: 6

Books list:
```

```
Enter your choice: 6
```

```
Books list:
```

```
1 | harry potter
```

```
2 | mocking bird
```

```
LIBRARY MENU
```

```
1. Add book information
```

```
2. Display book information
```

```
3. List all books of given author
```

```
4. List the title of specified book
```

```
5. List the count of books in the library
```

```
6. List the books in order of accession number
```

```
7. Exit
```

```
Enter your choice: 7
```

```
Exit
```

### 3- Tasks

- i- Declare a structure named employee that stores the employee id, name, salary and department.
- ii- Declare an array of 5 employees for the structure defined in part(i) . Also write statements to assign the following values to the employee [3].  
**Employee id = “Your\_roll\_no” salary = 30,000 and department = “IT dept”**
- iii- Write necessary statement to initialize all the elements of above array.
- iv- Write a function to take input in above array of struct employee.
- v- Write a function that prints the highest salaried person amongst the employees.
- vi- Write a function that search & display records of all those employees, whose salary is greater than 15000.
- vii- Write a function that search & display records of all those employees, who are working in Finance department.

```

#include <iostream>
#include <string>
using namespace std;

struct employee {
    string emp_id;
    string name;
    double salary;
    string department;
};

void Employees(employee e[], int size) {
    for (int i = 0; i < size; i++) {
        if (i == 3) {
            continue;
        }
        cout << "\nEnter details of employee " << i + 1 << endl;

        cout << "Employee ID: ";
        cin >> e[i].emp_id;

        cout << "Name: ";
        cin.ignore();
        getline(cin, e[i].name);

        cout << "Salary: ";
        cin >> e[i].salary;

        cout << "Department: ";
        cin.ignore();
        getline(cin, e[i].department);
    }
}

void highestSalary(employee e[], int size) {
    int max = 0;

    for (int i = 1; i < size; i++) {
        if (e[i].salary > e[max].salary) {
            max = i;
        }
    }
}
```

```
    cout << "\nEmployee with highest salary:\n";
    cout << "ID: " << e[max].emp_id << endl;
    cout << "Name: " << e[max].name << endl;
    cout << "Salary: " << e[max].salary << endl;
    cout << "Department: " << e[max].department << endl;
}

void salaryAbove15000(employee e[], int size) {
    cout << "\nEmployees with salary greater than 15000:\n";
    for (int i = 0; i < size; i++) {
        if (e[i].salary > 15000) {
            cout << "ID: " << e[i].emp_id
                << " Name: " << e[i].name
                << " Salary: " << e[i].salary
                << " Department: " << e[i].department << endl;
        }
    }
}

void finance(employee e[], int size) {
    cout << "\nEmployees in Finance Department:\n";
    for (int i = 0; i < size; i++) {
        if (e[i].department == "finance") {
            cout << "ID: " << e[i].emp_id
                << " Name: " << e[i].name
                << " Salary: " << e[i].salary << endl;
        }
    }
}

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023\n";

    employee emp[5];

    for (int i = 0; i < 5; i++) {
        emp[i].emp_id = "";
        emp[i].name = "";
        emp[i].salary = 0;
        emp[i].department = "";
    }
    emp[3].emp_id = "023";
    emp[3].name = "Ali";

    emp[3].name = "Ali";
    emp[3].salary = 30000;
    emp[3].department = "IT dept";

    Employees(emp, 5);
    highestSalary(emp, 5);
    salaryAbove15000(emp, 5);
    finance(emp, 5);
    return 0;
}
```

```
Microsoft Visual Studio Debug Console
Muhammad Ali
2025F-BIT-023
\

Enter details of employee 1
Employee ID: 1
Name: ahmed
Salary: 10000
Department: finance

Enter details of employee 2
Employee ID: 2
Name: ayan
Salary: 15000
Department: IT

Enter details of employee 3
Employee ID: 4
Name: anas
Salary: 20000
Department: CS

Enter details of employee 5
Employee ID: 5
Name: hamza
Salary: 25000
Department: finance

Employee with highest salary:
ID: 023
Name: Ali
Salary: 30000
Department: IT dept

Employees with salary greater than 15000:
ID: 4 Name: anas Salary: 20000 Department: CS
ID: 023 Name: Ali Salary: 30000 Department: IT dept
ID: 5 Name: hamza Salary: 25000 Department: finance

Employees in Finance Department:
ID: 1 Name: ahmed Salary: 10000
ID: 5 Name: hamza Salary: 25000
```

